

Day 5 — Mental-Math Speed Day

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will retest their Day 1 baseline problem (73×11) using the trick, complete a paired speed race, and finish Part B of the formative quiz.

Materials

- Day 1 baseline sheet (returned to each student).
- Stopwatch.
- `activities/activity-01-speed-race.md` worksheet.
- Formative quiz Part B (`quizzes/formative.md`).

Lesson flow

Warm-up (3 min)

- Quick chorus: 12 problems, mixed 2- and 3-digit $\times 11$. Cover the carry case.

Day 1 retest (5 min)

- Return Day 1 baseline sheet to each student.
- Re-solve 73×11 using the trick. Time yourself.
- Record:

Day 1 time (long multiplication): _____ sec
Day 5 time ($\times 11$ trick): _____ sec
Improvement: _____ %

- Most students will see $>70\%$ reduction (e.g. 25 sec $\rightarrow$ 5 sec).

Activity (20 min) — Paired speed race

See `activities/activity-01-speed-race.md` for the full set-up.

- Round 1 (5 min): Solo, 20 mixed-difficulty problems. Each problem on a flashcard.
- Round 2 (5 min): Paired. Partner reads problem; you answer aloud.
- Round 3 (5 min): Group chorus — teacher displays 10 problems; class shouts the answer.
- Round 4 (5 min): Quick-fire — teacher calls a 2-digit number; first student to call back the $\times 11$ product gets a tally mark. (Light competition; no cumulative score across days.)

Formative Quiz Part B (15 min)

- Distribute Part B.

- 8 questions, mixed 2- and 3-digit $\times 11$, plus 2 short-answer “explain why” items.

Wrap-up (2 min)

- Preview Day 6: “Tomorrow — the trick extends to $\times 12$, $\times 13$, ... all the way to $\times 19$. The middle-add step scales by a multiplier.”

Student-facing content

Speed Day

Today’s question: how much faster has your mental $\times 11$ gotten in 4 days?

Record your Day 5 retest time. Compare with Day 1. The gap is the proof of practice.

By end of class you should hit roughly: - 2-digit $\times 11$ (no-carry): under 2 seconds - 2-digit $\times 11$ (carry): under 4 seconds - 3-digit $\times 11$: under 8 seconds

Homework

- 25-problem mixed drill sheet. Solo, untimed. Target: 100% correct.

Differentiation

- **Tier up:** Try 4-digit $\times 11$ problems. Sample: 1234×11 , 2345×11 , 4567×11 . Use the right-to-left carry-cascade method.
- **Tier down:** Focus only on the 2-digit drill today. Build fluency before adding 3-digit.

Teacher notes

- The Day 1 $\rightarrow$ Day 5 contrast is the emotional payoff of the module. Make sure every student records it; the personal evidence drives motivation.
- Quick-fire (Round 4) should NOT escalate into pure competition. Use it lightly; rotate “first responder” so no single student dominates.
- Some students will plateau on the carry case. Pair them with a peer-coach in Round 2 specifically.

Citation(s) used in this lesson

- (No new sources today — applied practice of Days 1–4.)